Project Overview & Context:

The primary objective of this phase is to compile a structured, detailed technical report that documents the end-to-end implementation lifecycle of the “Auto Ticket Classification using Flow Designer” project. This documentation serves as an operational blueprint and a technical reference for administrators, developers, and evaluating faculty. It covers the initial design choices, technical requirements, platform configurations, and testing outcomes.

Technical Specifications & Architecture

The project was deployed on a cloud-based ServiceNow Personal Developer Instance (PDI) utilizing native ITSM capabilities. The automation pipeline bypasses legacy script-heavy methods by leveraging a low-code/no-code approach via Flow Designer.

System Ingestion: Captures active records upon creation on the Incident [incident] table.

Analysis Engine: Parses the Short description field using structured text condition matrices.

Routing Module: Executes data updates on backend record fields like Assignment group, Category, and Subcategory.

Workflow Configuration Breakdown

Trigger Initialization:

Set to execute asynchronously whenever a user registers a new support request.

Conditional Logic Evaluation:

Predefined criteria matrices inspect data inputs for key terms (e.g., password, network, hardware) to dynamically isolate user issues.

Targeted Action Processing:

Once conditions are satisfied, system actions rewrite target parameters, accurately assigning ownership to specific IT queues without human triage.

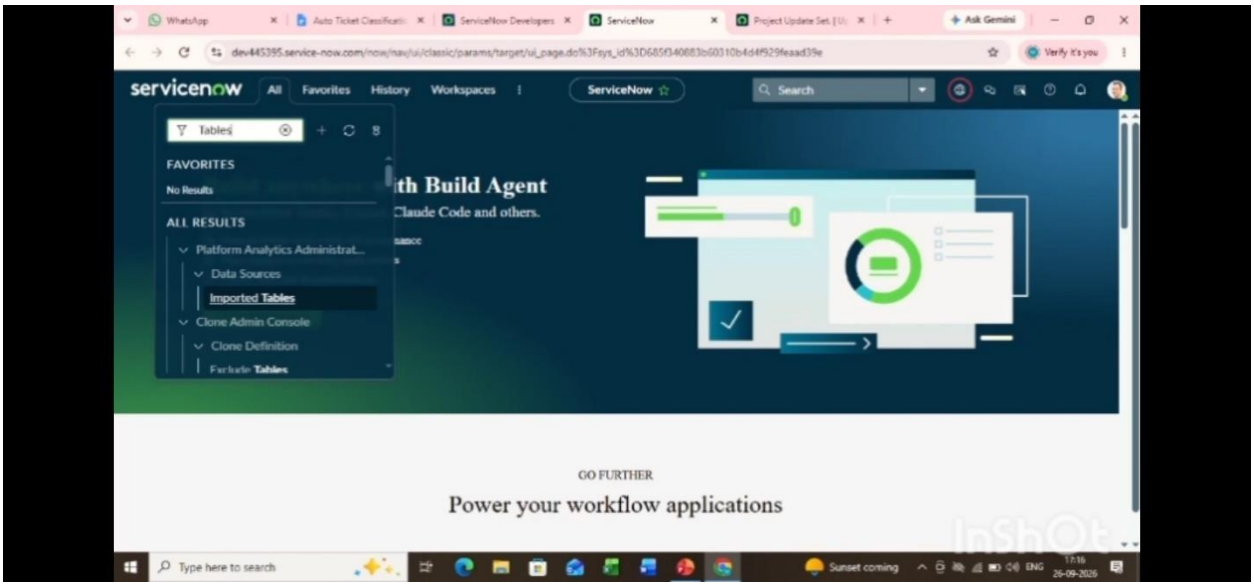
System Notification Module:

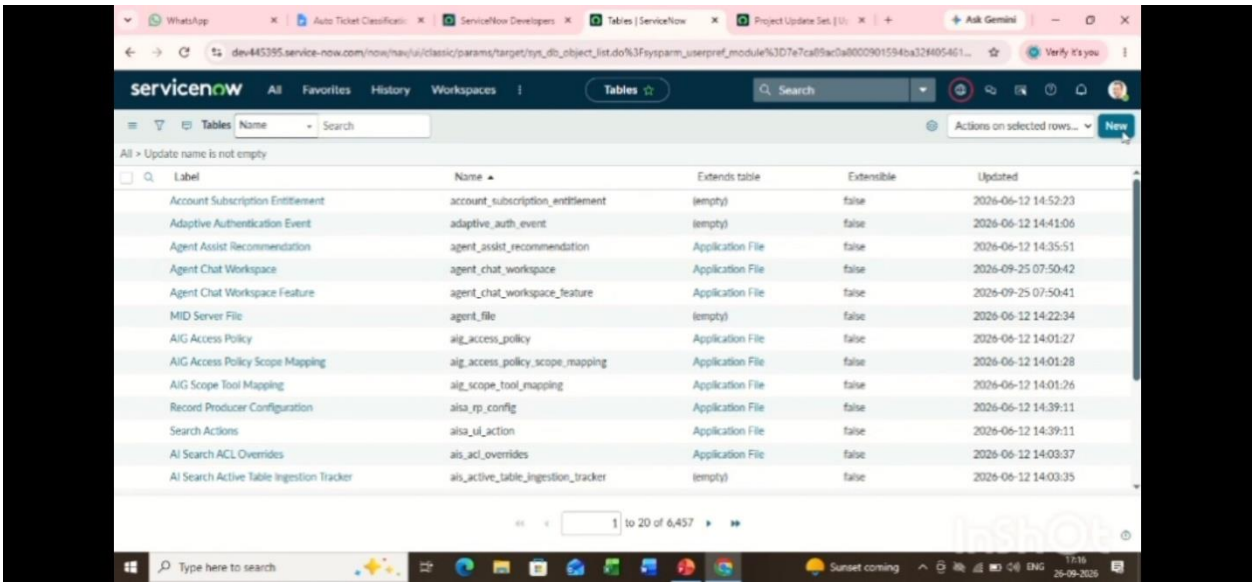
Operates an automated outbound communication setup that generates client confirmations, ensuring tracking and operational transparency.

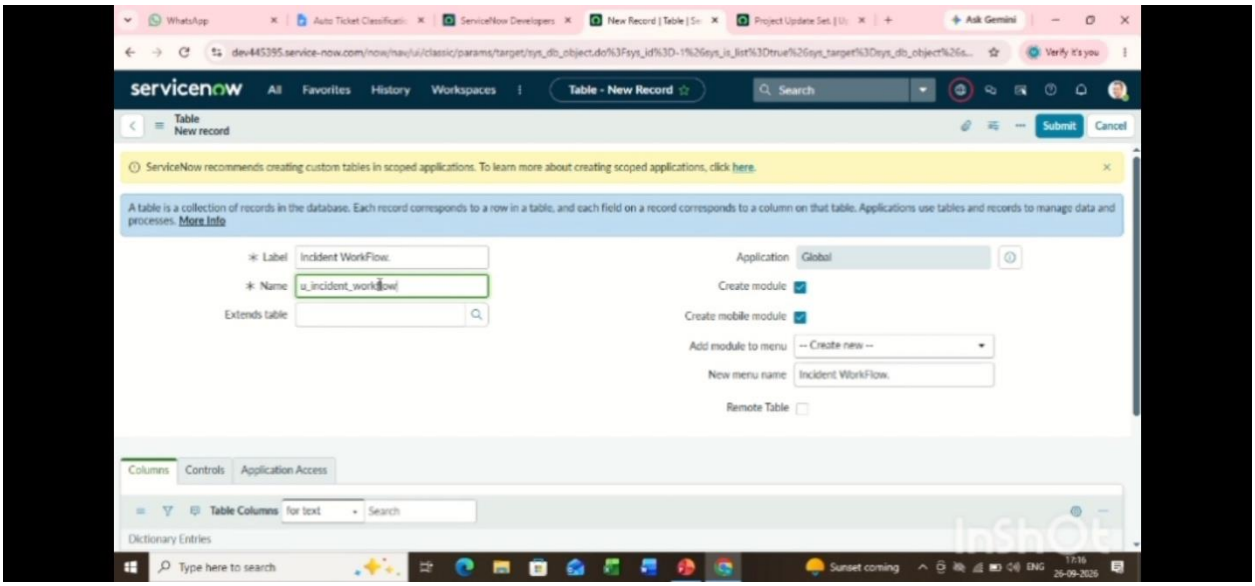
Testing, Validation & Verification Logs

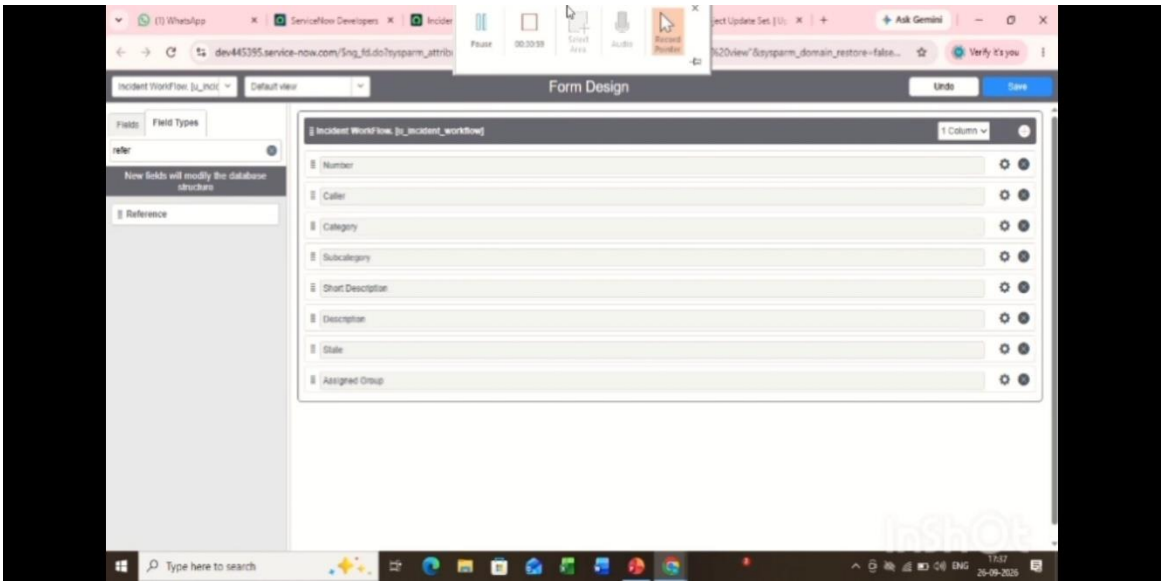
To guarantee system stability and verify routing logic under production scenarios, structured test cases were processed. The automatic execution engine successfully tracked and evaluated user text, routing all mock incidents correctly:

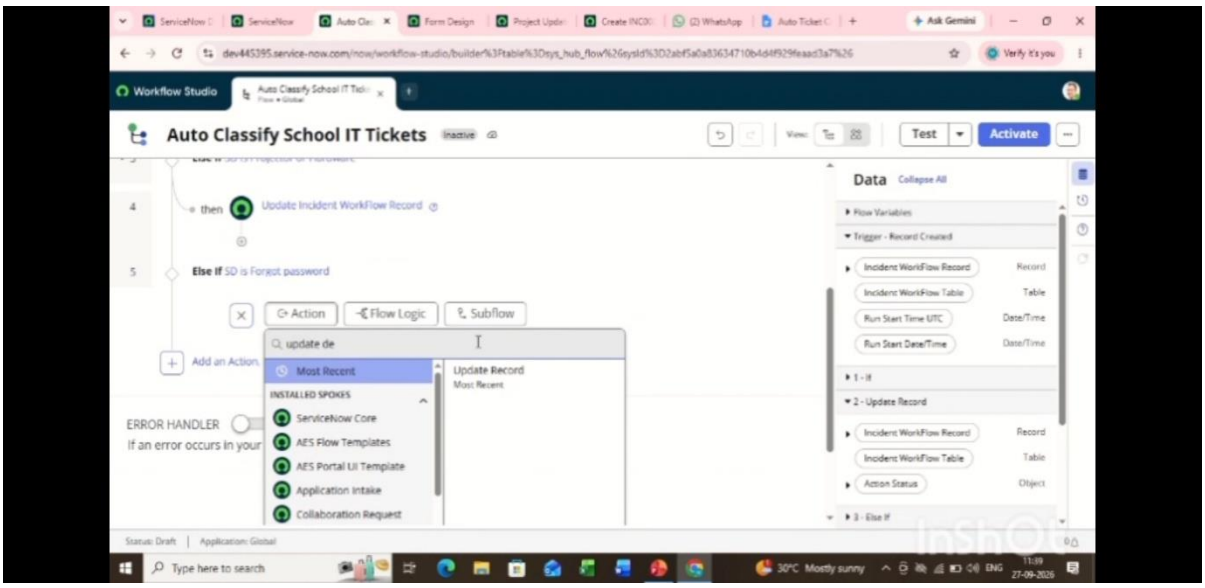
Test Case ID	Sample Input String	Expected Group	Actual Group	Status
TC-01	Cannot login to my student account	IT Helpdesk	IT Helpdesk	PASSED
TC-02	Lab room Wi-Fi is not connecting	Network Support	Network Support	PASSED
TC-03	Desktop screen is broken in library	Hardware Team	Hardware Team	PASSED




servicenow
All
Favorites
History
Workspaces
Tables
Search
Actions on selected rows...
New
All > Update name is not empty
Label
Name
Extends table
Extensible
Updated
Account Subscription Entitlement
account_subscription_entitlement
(empty)
false
2026-06-12 14:52:23
Adaptive Authentication Event
adaptive_auth_event
(empty)
false
2026-06-12 14:41:06
Agent Assist Recommendation
agent_assist_recommendation
Application File
false
2026-06-12 14:35:51
Agent Chat Workspace
agent_chat_workspace
Application File
false
2026-09-25 07:50:42
Agent Chat Workspace Feature
agent_chat_workspace_feature
Application File
false
2026-09-25 07:50:41
MID Server File
agent_file
(empty)
false
2026-06-12 14:22:34
AIG Access Policy
aig_access_policy
Application File
false
2026-06-12 14:01:27
AIG Access Policy Scope Mapping
aig_access_policy_scope_mapping
Application File
false
2026-06-12 14:01:28
AIG Scope Tool Mapping
aig_scope_tool_mapping
Application File
false
2026-06-12 14:01:26
Record Producer Configuration
aisa_rp_config
Application File
false
2026-06-12 14:39:11
Search Actions
aisa_ui_action
Application File
false
2026-06-12 14:39:11
AI Search ACL Overrides
ais_acl_overrides
Application File
false
2026-06-12 14:03:37
AI Search Active Table Ingestion Tracker
ais_active_table_ingestion_tracker
(empty)
false
2026-06-12 14:03:35
1 to 20 of 6,457
Type here to search
Sunset coming
ENG
17:16
26-09-2026


servicenow
All
Favorites
History
Workspaces
Table - New Record
Search
Table
New record
Submit
Cancel
ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click here.
A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. More Info
Label
Incident WorkFlow
Name
u_incident_workflow
Extends table
Application
Global
Create module
Create mobile module
Add module to menu
-- Create new --
New menu name
Incident WorkFlow
Remote Table
Columns
Controls
Application Access
Table Columns
for text
Search
Dictionary Entries
Type here to search
Sunset coming
ENG
17:16
26-09-2026


Form Design
Undo
Save
Fields
Field Types
refer
New fields will modify the database structure
Reference
Incident WorkFlow. [u_incident_workflow]
1 Column
Number
Caller
Category
Subcategory
Short Description
Description
State
Assigned Group
Type here to search


Workflow Studio
Auto Classify School IT Tickets
Inactive
Test
Activate
then
Update Incident WorkFlow Record
Else If SD is Forgot password
Action
Flow Logic
Subflow
update de
Most Recent
Update Record
INSTALLED SPOKES
ServiceNow Core
AES Flow Templates
AES Portal UI Template
Application Intake
Collaboration Request
ERROR HANDLER
If an error occurs in your
Add an Action
Data
Collapse All
Flow Variables
Trigger - Record Created
Incident WorkFlow Record
Incident WorkFlow Table
Run Start Time UTC
Run Start Date/Time
2 - Update Record
Action Status
Status: Draft
Application: Global
Type here to search

Project Outcomes & Technical Summary

Conclusion:

The automated workflow successfully replaces manual, time-consuming ticket triage with immediate cloud-based logic evaluation. The operational turnaround time for support distribution decreased from hours to real-time milliseconds, removing internal helpdesk dispatch bottlenecks.

Strategic Future Enhancements:

Integrating ServiceNow Predictive Intelligence to scale the routing system via Machine Learning text classification instead of static exact-match keyword clauses.

Setting up ServiceNow Virtual Agent chat nodes to attempt immediate self-service incident resolution prior to raising formal support records.